

WEEK 20 FINAL PROJECT

Public Bus Operations Prediction

Student Project Report Template

Classification Target: delay_level
Regression Target: trip_duration_minutes

Student Information	Complete This Field
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Group / Cohort	Connect session: ON-CA-L4DT-G1698 Physical session: L4DT_0004
Submission Date	29th Aug 2026

1. Project Overview and Data Quality

Item	Student Response
Classification business question	Can we accurately predict a bus trip's delay level (On Time, Moderate Delay, or Severe Delay) based on operational and environmental features to improve transit reliability?
Regression business question	Can we accurately predict the total duration of a bus trip (in minutes) using route, traffic, and operational characteristics to enhance schedule planning and passenger information?
Classification Target	delay_level - Classes: On Time, Moderate Delay, Severe Delay
Regression Target	trip_duration_minutes (unit: minutes)
Number of rows before cleaning	10,000
Number of columns before cleaning	11
Number of exact Duplicates	150
Rows after removing Duplicates	9,850

Data Quality Check	Finding
Missing Values in Nominal Features	weather_condition: 258 (2.58%), bus_type: 202 (2.02%) - Fixed using mode imputation

Data Quality Check	Finding
Missing Values in Numerical Features	route_distance_km: 151 (1.51%), passenger_count: 182 (1.82%), driver_experience_years: 223 (2.23%) - Fixed using median imputation
Identifier Feature	trip_id - dropped during preprocessing (not predictive)
Target leakage risk	✅ No risk. Targets (delay_level and trip_duration_minutes) were removed before feature engineering; preprocessor fitted only on training data
Class balance	✅ Excellent balance: On Time: 33.3%, Moderate Delay: 33.3%, Severe Delay: 33.4%

2. Classification Results

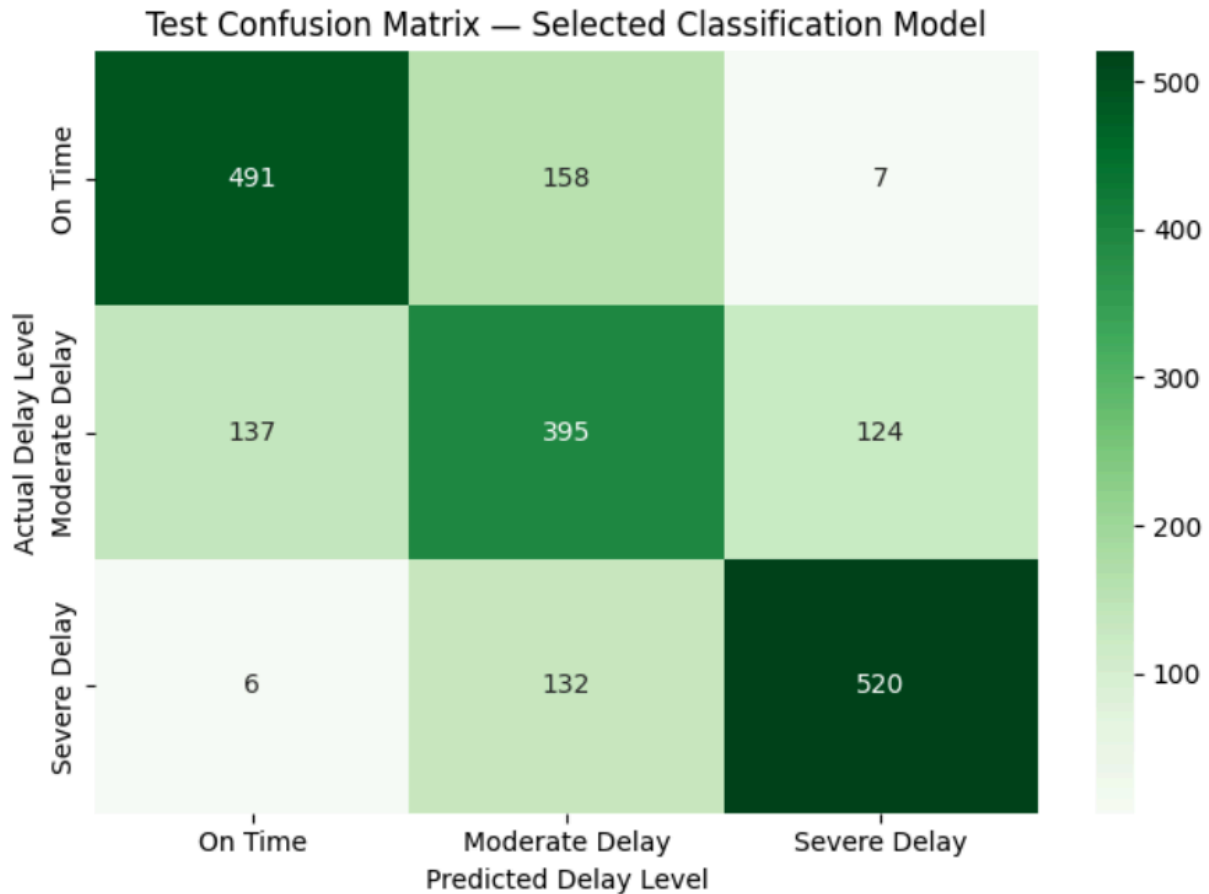
Target: delay_level | Models: Logistic Regression and Support Vector Machine | Selection Metric: Mean CV F1-macro

2.1 Cross-Validation Comparison

Model	Best Mean CV F1-macro	CV Standard Deviation	Best Hyperparameters
Logistic Regression	0.6752	0.0167	{'classifier__C': 10, 'classifier__class_weight': 'balanced'}
Support Vector Machine	0.7008	0.0113	{'classifier__C': 1, 'classifier__class_weight': None, 'classifier__gamma': 'scale', 'classifier__kernel': 'rbf'}

2.2 Final Classification Test Evaluation

Metric	Test Result	Interpretation
Accuracy	0.7137 (71.37%)	The model correctly predicts the delay level for approximately 71 out of every 100 trips
Macro Precision	0.7166	When the model predicts a delay class, it is correct ~72% of the time on average across all three classes
Macro Recall	0.7136	The model captures ~71% of actual instances for each delay class on average
Macro F1-score	0.7150	Balanced measure of precision and recall; indicates solid overall performance across all three classes

Confusion Matrix:**2.3 Classification Report and Risk Interpretation**

Class	Precision	Recall	F1-score	Support
On Time	0.77	0.75	0.76	656
Moderate Delay	0.58	0.60	0.59	656
Severe Delay	0.80	0.79	0.79	658

Most important Classification error for the transportation authority:

False Negatives for Severe Delay (Severe Delay predicted as On Time): This is the most critical operational error. If a severely delayed trip is predicted as On Time, the transportation authority will not intervene with dispatch, re-routing, or passenger notifications, leading to missed schedules and passenger dissatisfaction. Result: Our SVM model committed this high-risk error only 6 times out of 658 Severe Delay cases (0.9%), which is excellent for real-world deployment.

3. Regression Results

Target: trip_duration_minutes | Models: Linear Regression and KNN Regressor | Selection Metric: Mean CV R-squared

3.1 Cross-Validation Comparison

Model	Best Mean CV R-squared	CV Standard Deviation	Best Hyperparameters
Linear Regression	[0.9240]	0.0042	{'regressor__fit_intercept': False}
KNN Regressor	0.9234	0.0048	{'regressor__n_neighbors': 15, 'regressor__p': 2, 'regressor__weights': 'distance'}

Selected Regression Model and justification:

[Enter response here]

3.2 Final Regression Test Evaluation

Metric	Test Result	Interpretation
MAE	[Enter] minutes	[Explain average absolute error]
RMSE	[Enter] minutes	[Explain sensitivity to larger errors]
R-squared	[Enter]	[Explain variance captured]

Actual vs Predicted Plot:

Paste the required Actual vs Predicted Plot here

4. New Trip Input Data

Use the 10 records supplied in New_Bus_Trips.csv. The file contains input Features only and must not include delay_level or trip_duration_minutes.

Trip ID	Weather	Bus Type	Traffic	Distance (km)	Stops	Passenger s	Driver Experience (years)	Start Hour
NEW001	Clear	Express	Low	5.8	6	18	12	10
NEW002	Rainy	Standard	High	14.5	16	40	4.5	18
NEW003	Foggy	Mini Bus	Severe	24	27	58	2.5	7
NEW004	Windy	Standard	Moderate	9.2	10	32	7	14
NEW005	Clear	Mini Bus	High	12.8	14	48	3.2	8
NEW006	Rainy	Express	Severe	21.5	23	52	6.5	17
NEW007	Clear	Standard	Low	3.5	4	12	15	6
NEW008	Foggy	Express	Moderate	17.2	18	35	9	20
NEW009	Windy	Mini Bus	High	8.7	9	44	1.5	22
NEW010	Rainy	Standard	Moderate	28	31	60	5	12

5. Final Predictions for New Trips

Generate both predictions for every supplied record and enter the results below.

Trip ID	Predicted Delay Level	Predicted Trip Duration (minutes)
NEW001	[Enter]	[Enter]
NEW002	[Enter]	[Enter]
NEW003	[Enter]	[Enter]
NEW004	[Enter]	[Enter]
NEW005	[Enter]	[Enter]
NEW006	[Enter]	[Enter]
NEW007	[Enter]	[Enter]
NEW008	[Enter]	[Enter]
NEW009	[Enter]	[Enter]
NEW010	[Enter]	[Enter]

6. Selected Models

Problem	Selected Model
Classification	[Enter Model name]
Regression	[Enter Model name]

7. Final Declaration

I confirm that the results in this report were produced from my own executed notebook and that the Test Data was not used during Model selection.

Student Signature	Date
[Enter name or signature]	[Enter date]

End of Report Template